Case Study

Project: Document Management System (DMS)

Task 2 - Project Planning

Prepared by	Role
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1. Introduction

This case study describes the situation that led to the Document Management System project. The scenario was prepared by the team for this project.

2. Background

The case is about the administrative office of a university department. The office handles many types of documents every day: student files, official letters, meeting minutes, contracts, reports and forms. The office is managed by the Department Head, who is also the sponsor of the project.

The documents are kept in several places: paper folders in cabinets, personal computers, flash drives and email attachments. There is no single place where all documents are stored and no common rule for naming or filing them.

3. Problems

Problem	Effect on the department
Documents are hard to find	Staff lose time searching cabinets, computers and email.
Several versions of the same document	People do not know which version is the latest, which causes mistakes.
Paper documents are lost or damaged	Important papers cannot be recovered.
No control over who can see or change a document	Private documents may be seen by people who should not see them.
Approvals are done on paper	A document waits for a signature for days.
No record of changes	It is not possible to know who changed or deleted a document.

4. Stakeholders

Stakeholder	Interest in the system
Department Head (sponsor)	Wants control, security and faster approvals; approves the project.
Administrative staff	Main users; store, search and organize documents every day.
Faculty members	Need to find and share documents such as reports and minutes.
IT support	Installs, backs up and maintains the system.
Project team	Analyses and designs the system (Project Manager / Business Analyst, System Designer / UI/UX Designer, Documentation & QA).

5. Proposed Solution

A web-based Document Management System that stores documents in one central place, supports quick search, keeps a version history, controls access by user role, supports a simple approval process and records all activity in a log.

6. Planning Data

Before the analysis starts, the team planned its first four activities:

Activity	Description	Predecessors	Duration (days)
A	System Request Formulation	-	2
B	Feasibility Study (Economic & Technical)	-	4
C	Methodology Selection & Documentation	B	5
D	Time Estimation & Scheduling	A, C	3

7. Case Questions and Answers

Question 1: Why does the department need a Document Management System?

Because documents are scattered, versions are confused, papers are lost and there is no access control or history. A central system solves these problems and saves time.

Question 2: Which methodology should the team use?

Waterfall. The scope and deliverables are fixed, requirements are stable and the work is mainly documents (see the Methodology document).

Question 3: How long will the planning phase take and what is the critical path?

The planning phase takes 12 days. The critical path is B, C, D (4 + 5 + 3 = 12 days). Activity A has 7 days of slack.

Question 4: Is the project economically feasible?

Yes. The estimated first-year cost is $4,800 - $10,200 and the estimated annual benefit is $8,000 - $10,000+. With middle values, the payback period is about 10 months, the 3-year NPV is about $13,395 and the 3-year ROI is about 269% (see Lab 4).

Question 5: What are the main risks?

Risk	Response
Staff resist changing from paper to the new system	Training sessions and an easy-to-use interface.
Private documents are exposed	Role-based access, activity log and backups.
Scope grows during the project	Change control: changes are discussed at the weekly meeting and recorded.
A critical activity is late	Weekly check of the critical path and a one-day buffer.
Team members have limited time	Clear task ownership and a work-in-progress limit on the board.

8. Lessons Learned

- A clear problem statement and a system request make the rest of the planning easier.

- Finding the critical path early shows which activities must not be delayed.
- Choosing the methodology early (Waterfall) lets the team organize tasks, owners and deadlines.
- Economic analysis gives the sponsor a clear reason to approve the project.